

Tanmay Mirgal

Full-Stack Engineer | ML/DL Developer | AI & Computer Vision

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SUMMARY

- Full-Stack Engineer and ML/DL Developer with internship experience at Tata Advanced Systems Ltd. (TASL) and VigyBag, specialising in MERN applications, ML pipelines, and computer vision systems.
- Multi-hackathon winner, published researcher, and active open-source contributor with a track record of delivering production-ready systems independently.

TECHNICAL PROFICIENCIES

Languages & Frameworks: JavaScript (ES6+), TypeScript, Python, React.js, Next.js, Node.js, Express.js, FastAPI

Mobile & UI: React Native, Expo, Tailwind CSS, Shadcn/UI, Framer Motion, Bootstrap, Redux

Backend & Auth: REST APIs, WebSockets, WebRTC, BetterAuth, JWT, OAuth 2.0, Prisma, Drizzle, Mongoose

ML / AI / Computer Vision: TensorFlow, OpenCV, XGBoost, RAG, Groq AI, Deep Learning, Predictive Modeling, MLOps

Databases & Platforms: MongoDB, PostgreSQL, Redis, Convex, Supabase

Cloud & DevOps: AWS, Docker, CI/CD, Kubernetes, Nginx, Linux, Vercel, Git, GitHub

Tools & Methodologies: Postman, Jira, Bitbucket, Confluence, Agile, Scrum, Jest, Vite

WORK EXPERIENCE

VigyBag - Open-Source Software Engineer Intern

2024

Remote, India

- Contributed as an active developer on **VigyBag**, an open-source e-commerce platform bridging rural artisans with urban consumers across India (182+ GitHub stars, 419+ forks, 168+ contributors).
- Built and shipped React.js UI components and Node.js backend features for location-based product discovery and marketplace functionality.
- Worked on product search, category filtering, and pricing comparison across 10+ product categories, improving the vendor and buyer experience.
- Collaborated with a large open-source community; raised and resolved GitHub Issues, reviewed PRs, followed CI/CD and contribution guidelines.
- Stack: React.js, Redux, Node.js, Tailwind CSS, Vite, MongoDB, JavaScript, Vercel.

Tata Advanced Systems Ltd. (TASL) - Software Engineer Intern

2024 – 2025

Mumbai, India

- Engineered **CTS-71**, a confidential high-fidelity naval simulation platform replicating real-world Combat Management Systems for the Indian Navy, built on the MERN stack with WebSockets.
- Wrote automated test suites using Jest to ensure reliability and performance of mission-critical simulation modules.
- Collaborated with senior engineers in an Agile environment using Jira, Bitbucket, and Confluence for sprint planning and delivery.
- Received official offer letter and internship completion certificate from TASL upon successful delivery of all assigned modules.

PROJECTS

Samadhan — AI Civic Issue Reporting Platform

[Demo](#)

- Won 1st place at Internal Hackathon 2025 (Vidyalankar Polytechnic); built an AI-driven civic issue reporting ecosystem with real-time tracking using **Next.js, TypeScript, Convex, Expo, and Tailwind CSS**.
- Implemented cross-platform support with a React Native mobile app alongside the web interface for seamless citizen engagement.

Smart Meeter — AI Enterprise Meeting Platform

[Demo](#)

- Built an AI-powered enterprise meeting platform with custom WebRTC, real-time AI transcriptions, and Kanban workflow management using **React, FastAPI, TensorFlow, OpenCV, and Groq AI**.
- Integrated Groq AI for fast NLP-based meeting summaries and computer vision analytics for real-time video processing.

Finch AI — Intelligent Banking Recommendation Engine

[Demo](#)

- ML-driven banking recommendation engine serving personalised financial products to **500+ active users** using Next.js, Flask, XGBoost, and PostgreSQL.
- Deployed XGBoost classification models with full MLOps pipelines for continuous model retraining and production monitoring.

Cognix — Enterprise AI Assessment Mobile App

[Demo](#)

- Enterprise-grade AI Assessment & Analytics mobile app for clinical cognitive testing using **React Native, Express.js, MongoDB, RAG, and AWS**.
- Implemented Retrieval-Augmented Generation (RAG) for intelligent question generation, deployed on AWS with full MLOps pipelines.

EDUCATION

Vidyalankar Polytechnic | **Diploma in Computer Engineering** | 2023 – 2026

Mumbai, Maharashtra, India

CERTIFICATIONS

- TensorFlow for Deep Learning — Udemy | Aug 2025
- Complete AI & Machine Learning, Data Science BootCamp — Udemy | Aug 2025
- Use Generative AI for Software Development — IBM SkillsBuild (Student Ambassador Program) | Dec 2024
- Explore Text-to-Speech Using IBM Watsonx — IBM SkillsBuild (Student Ambassador Program) | Dec 2024
- DevOps Discover — Microsoft Student Ambassadors – Imagine Cup | 2024
- Introduction to Large Language Models — Google Cloud Skills Boost | Sep 2024
- Introduction to Responsible AI — Google Cloud Skills Boost | Sep 2024
- Contributor in GSSOC Open Source — GirlScript Summer of Code | Aug 2024
- Learn Advanced C++ Programming — Udemy | Sep 2024
- C++ in Programming in Hindi — Great Learning | Nov 2024
- Data Structures in C — Great Learning | Nov 2024
- Visual Graphics in C — Great Learning | Sep 2024
- Postman API Fundamentals Student Expert — Postman | 2024
- The Complete Python Bootcamp From Zero To Hero — Udemy | Nov 2023

ACHIEVEMENTS

- **Industrial Hackathon 2026 — Winner:** TATA | Built an enterprise-grade AI solution under competition conditions.
- **Internal Hackathon 2025 — Winner:** Vidyalankar Polytechnic | Built Samadhan, an AI civic platform.
- **Research Paper Published:** JAAFR Journal (2025) | Peer-reviewed AI and engineering research.
- **INNOVGENIUS IDEATHON Participant:** Tata Consultancy Services (2025) | National-level ideathon by TCS.
- **SPIT Hackathon 2025 Participant:** Sardar Patel Institute of Technology | Inter-college competitive hackathon.
- **Top Contributor — GSSOC 2024:** GirlScript Summer of Code | Recognised as a top open-source contributor.
- **Open-Source Contributor — VigyBag:** 168+ contributors, 419+ forks | Active merged contributor.